

Describing Data and Building Table 1

120 minutes · 6 self-check questions

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Summarising variables honestly: which measure of centre and spread fits which distribution, and how to assemble the participant characteristics table that opens almost every quantitative paper.

Learning objectives

- Choose an appropriate summary for each level of measurement
 - Judge distribution shape and pick mean or median accordingly
 - Compute and interpret standard deviation, IQR and confidence intervals
 - Build a Table 1 of participant characteristics
 - Decide when a p-value belongs in Table 1 and when it does not
 - Produce the table in SPSS and in R
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Matching the summary to the variable

- Nominal: counts and percentages only — a mean blood group is meaningless
 - Ordinal: median and interquartile range; means of Likert items are common but contested
 - Continuous and roughly symmetric: mean and standard deviation
 - Continuous and skewed: median and interquartile range
 - Always report the denominator alongside a percentage — 60% of 5 is not 60% of 500
 - Report percentages to at most one decimal place; more implies precision you do not have
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Judging the distribution

- Look at a histogram before choosing a summary; do not rely on a test alone
- Right skew is common for income, length of stay, biomarker concentrations and cost
- A mean far from the median signals skew
- Formal normality tests (Shapiro-Wilk) are oversensitive in large samples and underpowered in small ones
- In R: `hist(dat$x)` and `boxplot(dat$x)`; in SPSS, Analyze > Descriptive Statistics > Explore
- The question is not "is it exactly normal" but "is a mean a fair description of this distribution"

IN PRACTICE

Reporting a mean and standard deviation for a badly skewed variable is not wrong arithmetic — it is a misleading description. A mean income of GHS 4,200 with an SD of GHS 9,000 tells the reader almost nothing useful.

Spread and uncertainty

- Standard deviation describes the spread of the observations
- Standard error describes the precision of the estimate, and shrinks as the sample grows
- A confidence interval is the interval estimate that goes with a point estimate: $\text{mean} \pm 1.96 \times \text{SE}$ for a large sample
- Reporting SE instead of SD to make a figure look tidier misrepresents variability
- Interquartile range is the 25th to 75th percentile, and is robust to outliers
- Say which you are reporting: "mean (SD)" and "median (IQR)" are not interchangeable

CHOOSING A SUMMARY

Variable	Shape	Report	Example
Age	Roughly symmetric	Mean (SD)	42.3 (11.8) years
Income	Right skewed	Median (IQR)	GHS 1,800 (900-3,400)
Sex	Nominal	n (%)	312 (52.0%) female
Education	Ordinal	n (%) by level	104 (17.3%) tertiary
Length of stay	Right skewed	Median (IQR)	4 (2-9) days

Table 1

- One row per characteristic, columns for the whole sample and for each comparison group
- Give the denominator for every column, and note where it differs because of missing data
- Order rows sensibly: demographics, then clinical, then exposure
- Include the unit of measurement in the row label, not in every cell
- In R: the `tableone` or `gtsummary` package builds this directly from the data frame
- In SPSS: Analyze > Descriptive Statistics > Explore, split by group, then assemble

Should Table 1 contain p-values?

- In a randomised trial, no: any baseline difference is by definition due to chance, so testing it answers a question nobody asked
- CONSORT explicitly discourages significance testing of baseline characteristics
- In an observational study, a p-value can flag imbalance worth adjusting for, but the decision to adjust should be driven by subject knowledge, not by $p < 0.05$
- Standardised differences are a better measure of imbalance and do not depend on sample size
- Whatever you report, describe the comparison in the table footnote

Worked examples

Mean or median: an income variable

Monthly household income in a sample of 600: mean GHS 4,200, median GHS 1,800, standard deviation GHS 9,100, maximum GHS 180,000.

1. The mean is more than twice the median, which signals strong right skew.
2. The standard deviation exceeds the mean, so mean $\pm$ SD runs negative — impossible for income.
3. A histogram confirms it: most households below GHS 3,000, a long tail of a few very high earners.
4. Reporting "mean GHS 4,200 (SD 9,100)" describes almost nobody in the sample.
5. Report median GHS 1,800 (IQR 900 to 3,400), which describes the typical household.
6. If a mean is needed for a total or a cost calculation, report it separately and say why.

THE POINT

The arithmetic is fine either way; the description is not. When mean and median disagree sharply, the median is almost always the honest summary.

The p-value column that should not be there

A randomised trial submits Table 1 with a p-value column comparing baseline characteristics between arms. One row shows $p = 0.04$ for age, and the student wants to adjust for it.

1. Randomisation guarantees that any baseline difference arose by chance.
2. So the p-value tests a hypothesis known to be true, and 1 in 20 rows will be "significant" by construction.
3. CONSORT explicitly discourages significance testing of baseline characteristics.
4. The real question is not whether the difference is significant but whether it is large enough to matter.
5. A standardised difference answers that and does not depend on sample size.
6. Adjusting for a prognostic variable is fine — but decide that in advance from subject knowledge, not from a baseline p-value.

THE POINT

In a randomised trial, a baseline p-value answers a question nobody asked. In an observational study it can flag imbalance, but the decision to adjust still belongs to the conceptual framework, not the output.

Key terms

Standard deviation

The typical distance of observations from their mean; describes spread in the data.

Standard error

The standard deviation of an estimate; describes precision, and shrinks as n grows.

Interquartile range

The 25th to 75th percentile — a spread measure robust to outliers.

Skew

Asymmetry in a distribution, pulling the mean away from the median.

Table 1

The participant characteristics table summarising the sample, overall and by group.

Standardised difference

A measure of group imbalance independent of sample size, used instead of a p-value.

Activity

Build Table 1 for your own dataset, or a supplied one: overall column plus one column per group. For every row, justify in one line why you chose mean or median. Then decide whether a p-value column belongs, and defend it.

Further reading

- **Altman & Bland — Statistics Notes series, BMJ** — Short, free, and unmatched on describing data properly.
- **Assel et al. (2019) — "Guidelines for Reporting of Statistics"** — Concrete rules for tables and decimal places.
- **gtsummary package vignette (R)** — Produces a publication-ready Table 1 in a few lines.

Self-check questions

1. Household income in your sample has a mean of GHS 4,200 and a median of GHS 1,800. You should report:
 - a. Mean and standard deviation
 - b. Median and interquartile range, because the distribution is clearly right skewed
 - c. Either, they are equivalent
 - d. The mode
2. Standard error differs from standard deviation in that standard error:
 - a. Is always larger
 - b. Describes the precision of an estimate and shrinks as the sample grows
 - c. Can only be used for normal distributions
 - d. Is the square of the standard deviation
3. A reviewer objects to p-values in Table 1 of your randomised trial. They are:
 - a. Wrong — testing baseline balance is standard
 - b. Right — any baseline difference in a randomised trial arose by chance, so the test answers no useful question
 - c. Right only if the trial is small
 - d. Wrong, provided the test is non-parametric
4. Reporting "34.6% of participants" when the group contains 26 people is poor practice because:

- a. Percentages should never be used
 - b. One decimal place implies precision the denominator cannot support, and the count should be shown
 - c. 34.6% is mathematically impossible
 - d. SPSS cannot compute it
5. A Shapiro-Wilk test on 3,000 observations returns $p < 0.001$. The most sensible response is:
- a. Immediately switch to non-parametric tests
 - b. Look at a histogram — in large samples the test detects trivial departures from normality
 - c. Transform the variable regardless of its shape
 - d. Delete the outliers until it passes
6. For an ordinal education variable with five levels, the appropriate summary is:
- a. Mean and standard deviation
 - b. Counts and percentages for each level
 - c. Standard error only
 - d. Correlation coefficient
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Answers

- 1. **b.** A mean more than double the median indicates strong right skew, typical of income. The median describes the typical household; the mean is pulled up by a few high earners.
 - 2. **b.** SD describes variability among observations and does not systematically shrink with n . SE describes how precisely the mean is estimated and falls as the square root of n . Substituting SE for SD makes data look less variable than it is.
 - 3. **b.** Randomisation guarantees that baseline differences are chance. CONSORT discourages the practice; standardised differences are more informative if imbalance needs describing.
 - 4. **b.** With $n = 26$ each participant is 3.8 percentage points. Report "9 (35%)" so the reader sees what the percentage rests on.
 - 5. **b.** Normality tests are oversensitive in large samples: a departure too small to matter still returns a tiny p -value. Judge the shape visually and ask whether the mean is a fair description.
 - 6. **b.** The categories are ordered but the intervals between them are not equal, so a mean is not interpretable. Report n (%) per level, or a median category if a single summary is needed.
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